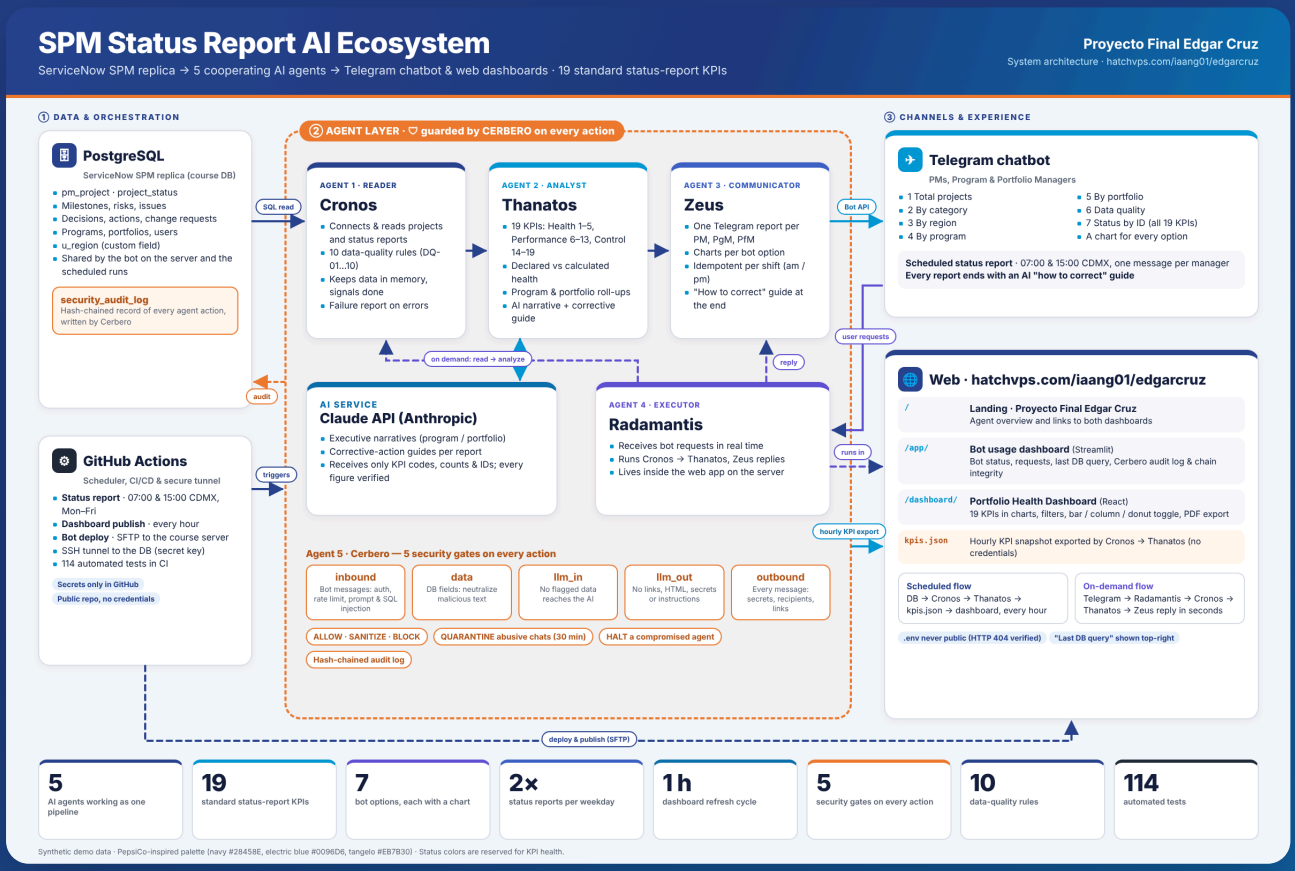
PROYECTO FINAL EDGAR CRUZ · SYSTEM DESCRIPTION

SPM Status Report AI Ecosystem

ServiceNow SPM replica → five cooperating AI agents → Telegram chatbot and web dashboards, delivering the 19 standard status-report KPIs with a security guardian on every action.



Site	hatchvps.com/iaang01/edgarcruz
Repository	github.com/richcruz1981/AI_M8_FinalAssignment_SPM_SR
Date	2026-10-18

This document describes the system as implemented on 2026-10-18. Anything not implemented is listed in section 13. Data shown in the system is synthetic demo data.

1. One-paragraph summary

A Python system reads project data from a PostgreSQL replica of ServiceNow SPM, computes the 19 standard status-report KPIs with five cooperating agents, and delivers them through three channels: (1) scheduled status reports on Telegram (07:00 and 15:00 America/Mexico_City, Monday–Friday), (2) an interactive Telegram chatbot answering 7 query types in real time, and (3) web dashboards on hatchvps.com (a React "Portfolio Health Dashboard" refreshed hourly and a Streamlit "bot usage" dashboard with the security audit log). A security agent (Cerbero) inspects every agent action and writes a hash-chained audit log. Claude (Anthropic API) writes executive narratives and an "how to correct" guide at the end of every report, receiving only KPI codes, counts and IDs.

5	19	7	5
AI agents	status-report KPIs	chatbot options	security gates

2. Components

ID	Component	Type	Responsibility	Code location
C1	**Cronos**	Agent 1 · Reader	Connects to the DB, reads projects, status reports and auxiliary tables, applies 10 data-quality rules (DQ-01...DQ-10), keeps data in memory, emits a completion signal, produces a failure report on errors	`src/spm_sr/cronos/`
C2	**Thanatos**	Agent 2 · Analyst	Computes KPIs 1–19, compares declared vs calculated health, builds program/portfolio roll-ups, requests AI narratives and AI corrective guides	`src/spm_sr/thanatos/`
C3	**Zeus**	Agent 3 · Communicator	Renders and sends Telegram messages (one per PM, Program Manager, Portfolio Manager), charts per bot option, idempotent per shift	`src/spm_sr/zeus/`
C4	**Radamantis**	Agent 4 · Executor	Receives chatbot requests in real time, runs Cronos → Thanatos, asks Zeus to reply; runs in a background thread inside the web app on the server	`src/spm_sr/radamantis/`

ID	Component	Type	Responsibility	Code location
C5	**Cerbero**	Agent 5 · Security	5 security gates on every agent action; quarantines abusive chats; halts compromised agents; hash-chained audit log	`src/spm_sr/cerbero/`
C6	Orchestrator	Pipeline	Cronos → Thanatos → Zeus with conditional edges on agent signals (COMPLETED / PARTIAL / FAILED); contract validation C1/C2/C3 (Pydantic)	`src/spm_sr/core/orchestrator.py`
C7	PostgreSQL	Data store	ServiceNow SPM replica + `security_audit_log` table	course server `alumnos-db`
C8	Claude API	External AI	Narratives and corrective guides	Anthropic Messages API
C9	Telegram Bot API	External channel	Scheduled reports and chatbot (long polling)	api.telegram.org
C10	GitHub Actions	Scheduler / CI/CD	Scheduled runs, hourly dashboard publish, bot deploy via SFTP, SSH tunnel to DB, tests	`.github/workflows/`
C11	Web landing	Static page	"Proyecto Final Edgar Cruz" section with links to both dashboards (PepsiCo palette)	`scripts/build_landing.py` → `/`
C12	Bot usage dashboard	Streamlit app	Bot status, requests, last DB query (top-right), summary, Cerbero audit log with chain integrity	`app.py` → `/app/`
C13	Portfolio Health Dashboard	React 18 + Recharts (Vite)	All 19 KPIs in charts, filters, bar/column/donut toggle, "Last DB query" (top-right), PDF export of the current view	`dashboard/` → `/dashboard/`
C14	KPI snapshot	Static JSON	Hourly export by Cronos → Thanatos (no credentials, no emails, no chat IDs)	`scripts/export_dashboard_data.py` → `/dashboard/data/kpis.json`

3. Diagram · Figure 1 · Ecosystem architecture

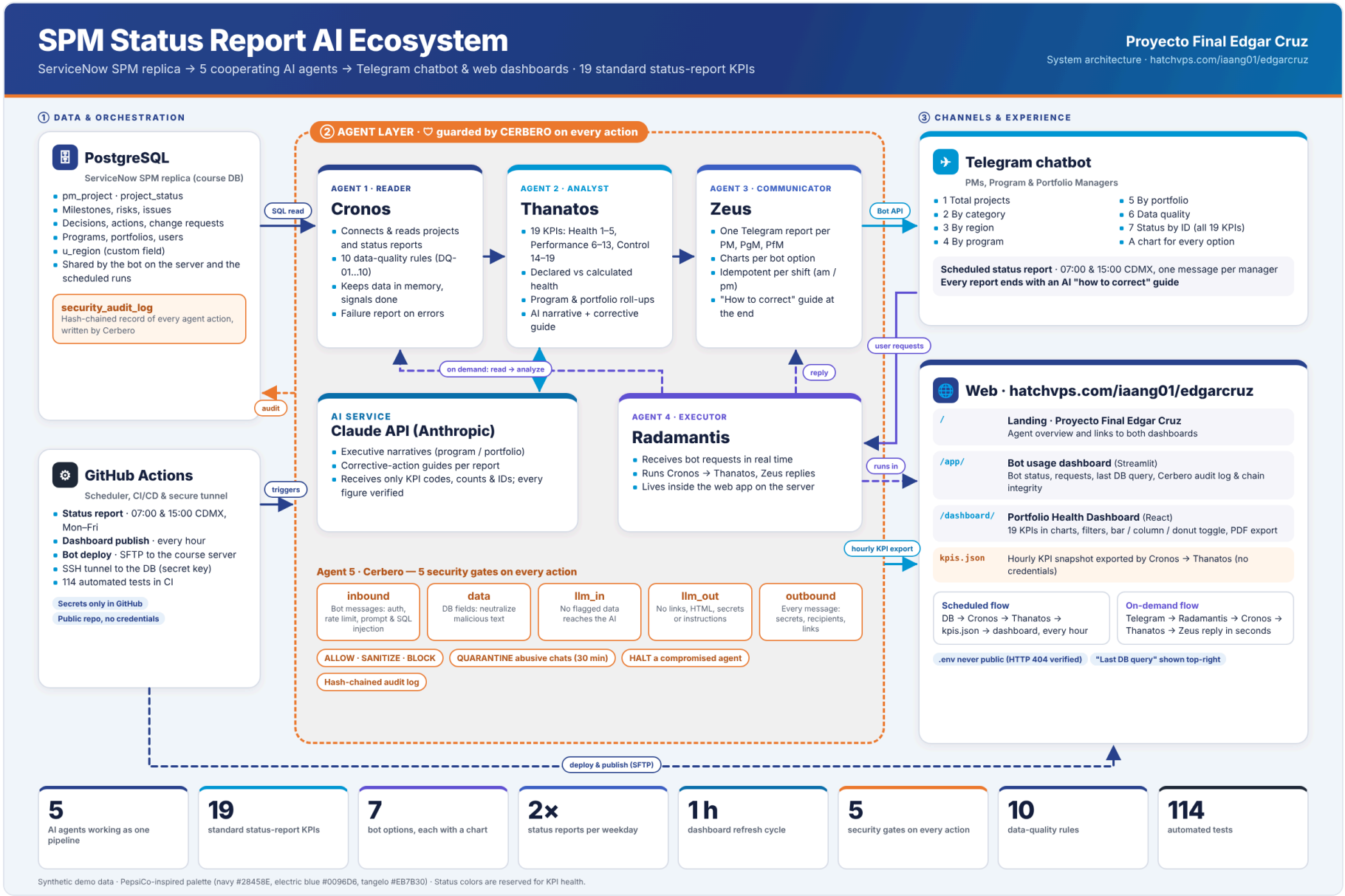


Figure 1 · Ecosystem architecture: data & orchestration, agent layer guarded by Cerbero, channels & experience.

4. Data flows

Flow	Trigger	Path	Output
F1 Scheduled status report	GitHub Actions cron `0 13 * * 1-5` and `0 21 * * 1-5` (07:00 / 15:00 CDMX)	SSH tunnel → Cronos → Thanatos (+ Claude) → Zeus → Telegram	One message per manager, ending with an AI corrective guide; idempotent per date + shift (`SR_SLOT` = am / pm / manual)
F2 Chatbot request	User message on Telegram	Telegram → Cerbero inbound → Radamantis → Cronos (cached 60 s) → Thanatos → Claude guide → Zeus (outbound gate) → Telegram	Text + chart (PNG) + AI guide
F3 Dashboard refresh	GitHub Actions every hour (`7 * * * *`)	SSH tunnel → Cronos → Thanatos → `kpis.json` → Vite build → SFTP to `/www/dashboard/`	Portfolio Health Dashboard with fresh data
F4 Bot deploy	Manual workflow	Package `spm_sr.zip` + `app.py` + JSON config + `.env` → SFTP	Server restarts `app.py` , bot thread starts (long polling)
F5 Security audit	Every agent action	Cerbero gate → `security_audit_log` (hash chain per session)	Visible in `/app/` (masked chat IDs, reason codes, no content)

5. KPIs (19)

Group	KPIs	Examples of fields
Health (1–5)	Overall, Schedule, Cost, Resources, Scope — calculated vs PM-declared (Consistent / Optimistic / Conservative / Not comparable)	Green / Amber / Red / No data
Performance (6–13)	% complete vs expected (linear), planned/actual start and end, end variance, planned vs actual cost, CPI, budget used, planned vs actual effort	`pctComplete` , `cpi` , `budgetUsedPct`
Control (14–19)	Milestones (done/overdue), open risks (high), open issues, pending decisions, pending/overdue actions, pending change requests	counts per project, summed in roll-ups

Data-quality rules: DQ-01 missing status report · DQ-02 stale status report · DQ-03 invalid % complete · DQ-04 invalid/empty planned dates · DQ-05 progress without actual start · DQ-06 missing planned cost or effort · DQ-07 no PM or PM without email · DQ-08 empty declared health · DQ-09 duplicate record · DQ-10 missing region.

6. Chatbot (Telegram) interface

Option	Command	Answer
1	`/total`	Total, active, closed/cancelled, analyzed, calculated health distribution
2	`/categories`	Open / In progress / On hold / Closed / Cancelled with raw states
3	`/regions`	Projects and active projects per region (custom field `u_region`)

Option	Command	Answer
4	`/programs`	Projects and health per program
5	`/portfolios`	Projects and health per portfolio
6	`/dataquality`	Projects with data-quality findings by rule
7	`/status <ID>` or just the ID	All 19 KPIs for a project (`PRJ…`), program (`PRG…`) or portfolio (`PFL…`)

Every option returns a chart and ends with "🧭 AI guide · how to correct" (or a labeled rule-based guide if AI is unavailable). Spanish command aliases still work. Only allow-listed chat IDs get data.

7. AI usage and controls

Use	Input sent to Claude	Validation	Fallback
Executive narrative (programs/portfolios first, ≤60 calls per run)	Roll-up / project KPI JSON	Every number must exist in the input; Cerbero `llm_out` (no links, HTML, secrets, instructions)	Deterministic template
Corrective guide (one per report / bot answer, cached by situation signature)	Only situation codes, labels, counts and validated IDs — no free text from the DB	Same as above	Rule-based playbook, labeled as such

The system prompt declares the JSON untrusted and forbids following instructions inside it.

8. Security (Cerbero)

Gate	Agent	Checks	Action
inbound	Radamantis	Authorization, ≤300 chars, ≤12 requests/min, control characters, prompt/SQL/command/HTML injection	BLOCK; 3 blocks in 10 min → 30-min quarantine (no replies)
data	Cronos	Every text field from the DB	Malicious field → `[removed by security]` , record flagged; >20 % flagged → HALT run
llm_in	Thanatos	Flagged records never reach the AI	Template instead
llm_out	Thanatos	AI output: links, HTML, secrets, instructions, length	BLOCK → template
outbound	Zeus	Secrets (tokens, API keys, DB URLs, private keys), recipient allow-list, disallowed HTML, foreign links	Secret or foreign recipient → HALT Zeus; links/HTML stripped

Audit record fields: `ts, source, session, agent, action, gate, decision, severity, reasons, subject (masked), content_sha256, excerpt (blocked items only), prev_hash, hash` . Other controls: secrets only in GitHub Actions secrets and the server `.env` (verified not public, HTTP 404); public repository contains no credentials (CI test); workflow inputs passed via environment variables and validated.

9. Deployment and runtime constraints

Constraint	Design decision
Course server keeps only `app.py` alive; other `.py` files run as 150-second scripts without libraries	Bot runs in a thread inside `app.py`; package shipped as `spm_sr.zip`; config shipped as JSON
Every file under `/www` is public	No log or audit files on the server; audit goes to the DB
SSH (port 22) unreachable from the authoring environment	All SFTP/tunnel steps run in GitHub Actions with a secret key
DB limit: 8 connections, 60 s per query	Cached reads (60 s) in the bot; audit writer pool of 1

10. Web pages

Path	Content
`/`	Original hosting welcome page restyled with the PepsiCo palette, with the "Proyecto Final Edgar Cruz" section first: agent description and links to both dashboards
`/app/`	Bot usage dashboard: status (listening/created/stopped), requests, errors, authorized chats, last DB query (top-right), summary tables, Cerbero audit log (24 h counters, chain integrity, filterable table)
`/dashboard/`	Portfolio Health Dashboard: 7 summary tiles, sections Health / Performance / Control / Breakdown / Data quality / Projects, filters (portfolio, program, region, category, health, PM, search, active/all), chart-type toggle, project drawer with all 19 KPIs, "Download PDF" (A4 landscape, page breaks between cards)

11. Visual identity

PepsiCo-inspired palette (secondary source, Coloropedia): navy `#28458E`, electric blue `#0096D6`, tangelo `#EB7B30`. Chart series `#3D63C4`, `#EB7B30`, `#0096D6`, `#6250D6`, `#E87BA4`, other `#A7AEBB`. Status colors are reserved: Green `#00984A`, Amber `#E0A000`, Red `#DF002C`, No data `#A7AEBB`. Font: Inter.

12. Key numbers

Metric	Value
Agents	5 (Cronos, Thanatos, Zeus, Radamantis, Cerbero)
KPIs	19
Bot options	7, each with a chart
Scheduled status reports	2 per weekday
Dashboard refresh	hourly
Security gates	5
Data-quality rules	10

Metric	Value
Automated tests	114 (at 2026-10-18)
Demo dataset	150 projects, 129 active, ~69 recipients per run

13. Not implemented / limitations

- Email channel (removed: no SMTP server available).
- Real-time DB streaming to the dashboard (hourly snapshot instead).
- Quarantine state persists only in the bot process memory (reset on server restart).
- Security rules are pattern-based: they reduce risk, they do not eliminate it.
- Data is synthetic; record names (projects, regions, people) are stored in Spanish.

14. Glossary

SPM: Strategic Portfolio Management (ServiceNow) · SR: Status Report · PM / PgM / PfM: Project / Program / Portfolio Manager · CPI: Cost Performance Index (EV/AC) · DQ: data quality · HALT: Cerbero stops an agent action · CDMX: America/Mexico_City (UTC-6).